Sidharth Bhagat

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Software Development Engineer
Software engineering graduate candidate building reliable APIs, distributed systems and accessible applications. Hands-on Python, C++, SQL and full-stack development, supported by crash testing, profiling and algorithmic problem solving.

EDUCATION

Indian Institute of Technology Jodhpur 2023–2027
B.Tech, Electrical Engineering | CGPA: 7.88/10 | Departmental rank 3 at end of first year.

EXPERIENCE

Software Engineer Intern – Data Engineering | ICICI Bank, Audit Intelligence Unit Summer 2026

- Built AIU DataVerse, a Python platform for heterogeneous financial data, with 9-phase Bayesian delimiter detection and 23-stage type recovery for automated schema inference.
- Migrated Pandas processing to Polars LazyFrame with SHA-256 content-addressed Parquet caching; measured 1,673x profiling and 12.91x CSV-parsing speedups on internship benchmark workloads.
- Implemented regression modeling, MinHash/LSH deduplication and population stability index (PSI) drift detection; audited the codebase and remediated 6 defects.

Software Development Fellow | SoLA, Comic Studies Summer 2025

- Built a comic-to-audio pipeline integrating computer vision, OCR, dialogue extraction and speech synthesis; processed 4 pages/minute in the fellowship workflow.
- Implemented speaker attribution for up to 5 characters and emotion-conditioned narration, preserving dialogue identity through the audio pipeline.

Undergraduate Research Intern | IIT Jodhpur, CS Department Summer 2024

- Benchmarked meta-heuristic and approximation approaches to combinatorial optimization with Prof. Tanmay Nitin Inamdar; achieved a 40% improvement in solution quality in the studied scenarios.

PROJECTS

WARDEN | **Distributed systems**

- Built authenticated FastAPI services with PostgreSQL, Redis Streams and transactional outboxes; recovered 500/500 jobs across 12 worker kills with no lost jobs or duplicate durable effects in crash tests.
- Implemented a typed rule language with a Pratt parser and RE2, exercised by 60,000 fuzz cases; added signed webhooks, retries, circuit breakers and dead-letter queues.
- Tuned PostgreSQL queries with indexes and EXPLAIN ANALYZE, reducing execution time by up to 99.6% on 150,002 fixture rows; added OpenTelemetry, Prometheus and Grafana observability.

Panel | **Accessible comic reader**

- Built a React/TypeScript reader and FastAPI/SQLite backend combining PDF extraction, EasyOCR and OpenCLIP embeddings with editable character casting and cross-page identity merging.
- Integrated Kokoro/ONNX speech with eSpeak fallback, cached audio, 28 English voice definitions and 13 prosody styles; added transcript controls, reading progress and accessibility preferences.

EncryptiView | **Document workflows**

- Implemented Django REST APIs and a Flutter client for document upload, marketplace browsing and owner-approved access requests, using JWT authentication and role-based permission checks.
- Rendered permitted PDF pages with PyMuPDF watermarks and added extracted-text narration through gTTS; connected secure client token storage to remote viewing workflows.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, TypeScript; data structures, algorithms, OOP, concurrency.

Backend and data: FastAPI, Django REST Framework, PostgreSQL, Redis Streams, SQLite, Polars, Parquet.

Delivery and interfaces: Docker, Git, Linux, OpenTelemetry, Prometheus, Grafana, React, Flutter.

ACHIEVEMENTS AND LEADERSHIP

- LeetCode Guardian (top 1.02%); CodeChef 5-star; 800+ problems solved.
- Top 1.5% among 1.2M+ JEE candidates; Assistant Head, Varchas 2025 and Prometeo 2025, IIT Jodhpur.